" If You Want To Manage It, Measure It "

IOTAFLOW
SYSTEMS PVT. LTD

LoRaWAN®

Inline Ultrasonic

V1 Robust Series

BATTERY-OPERATED
ULTRASONIC WATER METER

www.iotaflow.com

Frugal Solutions By Innovation

Keeping the Indian tradition of frugal invention alive, IOTAFLOW introduces a water meter which addresses a long standing problem of sand coming with raw water - a real headache for maintenance team.
Sand coming with water is enemy of moving parts in Woltman water meters. Here we provide you a water meter which is not only without any moving parts it is economical as well almost at the price of Woltman water meter.
This flow meter also helps you measure DM water, Hot water where Electromagnetic flow meter face problems we can offer these water meters with MS / CI / SS304 / SS316 body material to make it most versatile flow meter and you can use it for Raw water, Treated water, DM water, Hot water.

Now available with In-built LoRaWan Feature !!

Information Powerhouse

It has almost everything in display be it Flow Rate, Totalizer - Forward, Reverse, Net, No of hours meter was installed, worked.
Archives have record of hourly reading for 1440 Hours,
Daily reading for 460 days and monthly reading for 48 Months.
Communication options include Impulse, RS485, Mod bus Wired, Mod bus Wireless

NO PROBLEM Meter

- Outdoor installation - NO PROBLEM - IP67 protection
- No Power - NO PROBLEM - 10 years battery life
- Low Flow rate - NO PROBLEM - range-ability of 1:125 (Qmin:Qn)
- Communication - NO PROBLEM - lot of options include Impulse, RS485, Mod bus Wired, Mod bus Wireless, GSM/GPRS, NFC
- Sand / Foreign particles coming with water - NO PROBLEM - Through anything at this meter and it will work tirelessly.

Features

- Independent of Power failure and Power-off;
- Independent of water Pollution Level (filters Not Required);
- No Pressure Loss in pipeline;
- High Stability of Measurement;
- Electronic unit positioned directly on spool piece;
- LCD on/off button for battery charge saving;
- Additional options: Archives, Galvanic isolation for Pulse output, LCD back light, cable gland for Pulse Output ;
- Easy Commissioning;

Application

- Residential water consumption metering (cold and Hot tap water);
- Water consumption metering at industrial enterprises;
- Water metering at pipeline Control Points;
- Prepayment systems;
- Irrigation.

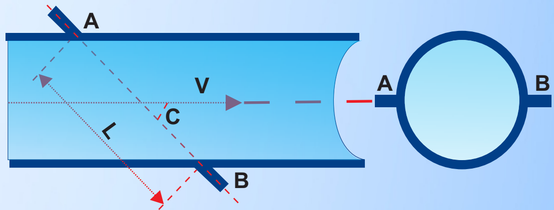
Technology

A typical transit-time flow measurement system utilizes two ultrasonic transducers that function as both ultrasonic transmitter and receiver. The flow meter operates by alternately transmitting and receiving a burst of sound energy between the two transducers and measuring the transit time that it takes for sound to travel between the two transducers. The difference in the transit time measured is directly and exactly related to the velocity of the liquid in the pipe.

To be more precise, let's assume that T down is the transit-time (or time-of-flight) of a sound pulse traveling from the upstream transducer A to the downstream transducer B, and Tup is the transit-time from the opposite direction, B to A. The following equations hold:

T down = (D / sin ø) / (c + V*cos ø), (1)

Tup = (D / sin ø) / (c - V*cos ø), (2)



Where c is the sound speed in the liquid, D is the pipe diameter and V is the flow velocity averaged over the sound path. Solving the above equations leads to

V = (D / sin 2ø) * ΔT / (Tup * T down), (3)

Where ΔT = Tup - T down. Therefore, by accurately measuring the upstream and downstream transit-time Tup and T down, we are able to obtain the flow velocity V. Subsequently, the flow rate is calculated as following,

Q = K *A* V, (4)

Where A is the inner cross-section area of the pipe and K is the instrument coefficient. Usually, K is determined through calibration.

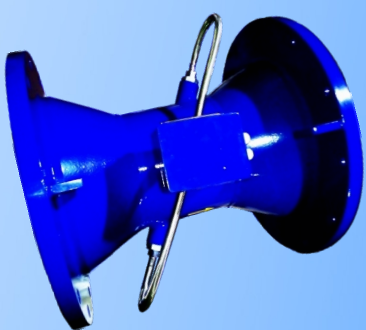
From equations (3) and (4), we see that the measurement results, V and Q, are independent of fluid properties, pressure, temperature, pipe materials, etc. The sound speed term does not appear in the final equations. These characteristics, plus large turn-down ratio, no pressure drop, no moving parts, no disturbance to the flow and many other features, make ultrasonic transit-time flow meter extremely attractive.

Specification

Parameter	Value
Nominal Diameter Of The Pipeline, DN	32, 40, 50, 65, 80, 100, 125, 150, 200, 250, 300, 350, 400
Accuracy, %	± 2.0
Flow velocity range, m/s	0.02 - 5
Fluid temperature range, °C:	50 / 130
Ambient temperature range, °C	5 - 65
Maximum pipeline pressure, MPa / bar	2.5 / 25
Protection	IP67
Compliance	ISO 4064
Measurement data logging, number of records: - hourly log - daily log - monthly log	 1440 460 48
Lithium battery life, Yrs.	10

The Basic Technical Characteristics

Diameter of conditional pass of the pipeline, DN								
50	65	80	100	125	150	200	250	300
The most smaller measured average volume expense, Qmin, m³/h								
0.28	0.48	0.73	1.13	1.17	2.55	4.53	7.08	10.19
The transitive measured average volume expense, Qtrans, m³/h								
1.40	2.40	3.60	5.70	8.80	12.7	22.6	35.4	50.9
The greatest measured average volume expense, Qmax, m³/h								
35.4	59.8	90.6	141.5	221.1	318.4	566.0	884.4	1273.5



Outputs

- LCD;
- Universal output;
- RS-485 interface (ModBus);
- Wired or wireless M-Bus interface – on request.

LoRaConnect 11

LoRa/RF Module Specifications

- RF output power up-to +22 dBm.
- It supports LoRa® Point to Point communications as well as LoRaWAN® protocol.
- Built-in EEPROM, data kept unchanged even powered off.
- Small size (21*18 mm)
- Wide range of working voltage 3V to 3.7 V
- Sensitivity -137dBm
- Wide range of temperatures -40°C to +85°C.

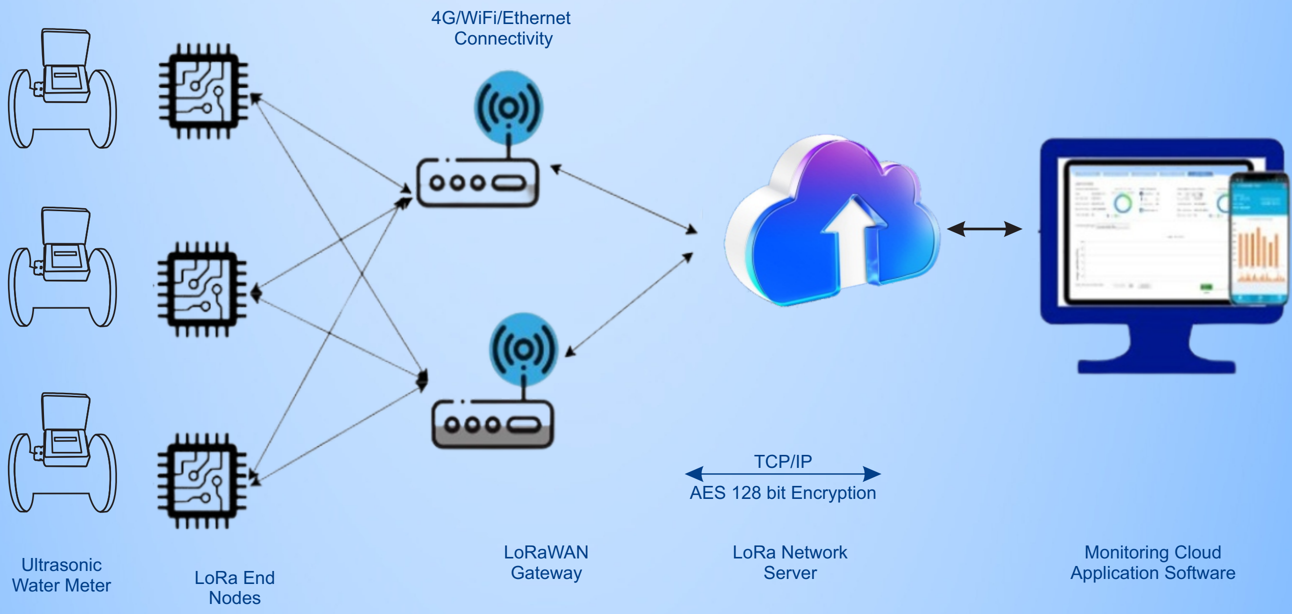
Applications

Typical applications for this Gateway include smart metering, wearables, tracking, M2M and internet of things (IoT) edge nodes.

The Gateway's applications are as following -

- Automated Meters Reading
- Industrial Monitoring and Control
- Long Range Irrigation Systems

Network Architecture of the Entire System

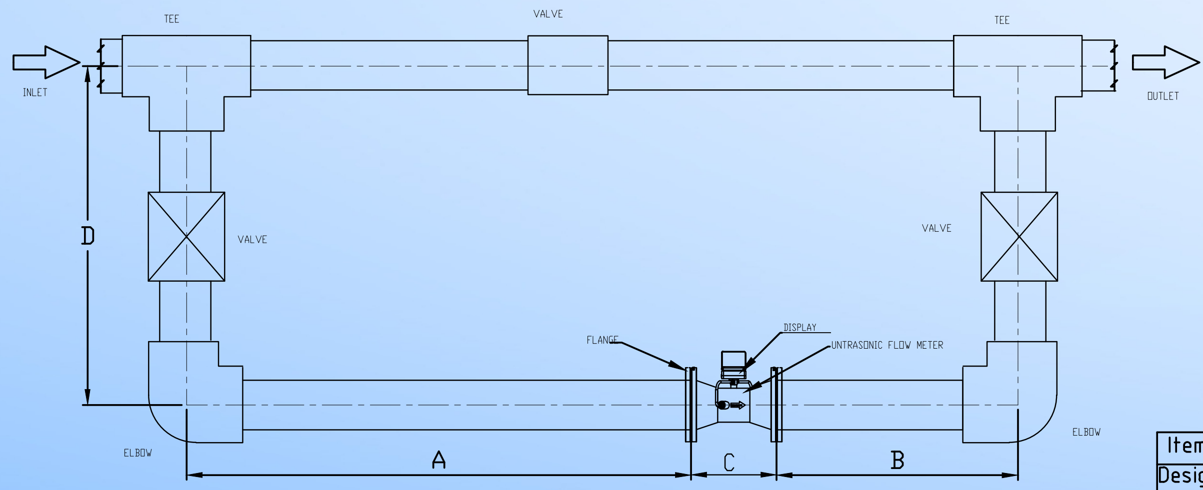


Installation Diagram

Dimension details				
Size	A	B	C	D
25	250	125	300	480
32	320	160	300	480
40	400	200	200	500
50	500	250	200	500
65	650	325	200	520
80	800	400	200	530
100	1000	500	250	550
125	1250	625	250	580
150	1500	750	300	610
200 Low Flow	2000	1000	350	660
200 High Flow	2000	1000	450	660
250	2500	1250	450	700
300	3000	1500	500	800
350	3500	1750	500	850
400	4000	2000	600	1000

Flange Details								
size	OD Ø	PCD Ø	Thk.(T)	No of hole(n)	Hole Ø	RF Ø	RF Thk.(f)	Flange std.
25	115	85	16	4	14	68	2	PN16
32	140	100	16	4	18	78	2	PN16
40	150	110	16	4	18	88	3	PN16
50	165	125	18	4	18	102	3	PN16
65	185	145	18	4	18	122	3	PN16
80	200	160	20	8	18	138	3	PN16
100	220	180	20	8	18	158	3	PN16
125	250	210	22	8	18	188	3	PN16
150	285	240	22	8	22	212	3	PN16
200	340	295	24	12	22	268	3	PN16
250	405	355	26	12	26	320	3	PN16
300	460	410	28	12	26	378	4	PN16
350	520	470	30	16	26	438	4	PN16
400	580	525	32	16	30	490	4	PN16

Note:- All dimension are in mm



Itemref	QTY:-	Material:-....	Date:-14/10/22	Reference:
Designed by Aman Saha	Checked by Pawan Arya	Approved by Pankaj Malik	File Name:-	Part No.:.......
IOTAFLOW SYSTEMS PVT.LTD. C-2/13, Mayapuri Industrial Area, Phase - II, New Delhi - 110 064 India.		Part Name:- Installation drawing UFM DN25 to DN400		
		Revision:-1.01	Scale:-nts	Sheet:-

	Inline UFM-	XXX	X	X	X	X	X	X		X	X	X	X	X	X	X
Size	UFM 32 Mm	32														
	UFM 40 Mm	40														
	UFM 50 Mm	50														
	UFM 65 Mm	65														
	UFM 80 Mm	80														
	UFM 100 Mm	100														
	UFM 125 Mm	125														
	UFM 150 Mm	150														
	UFM 200 Mm	200														
	UFM 250 Mm	250														
	UFM 300 Mm	300														
Mounting	UFM Inline		1													
Flow Tube Material	FMT CI / MS			M												
	FMT SS304			S												
	FMT SS316			F												
Flow Sensor Protect	FMT IP67				IP67											
	FMT IP68				IP68											
Structure	UFM Integral					1										
Process Connection	UFM MS / CI Flange						M									
	UFM SS 304 Flange						F									
	UFM SS 316 Flange						G									
	UFM SMS Union						S									
	UFM TC End (SS 316)						T									
Pressure Rating	UFM 16 Bar							B								
	UFM 25 Bar							C								
Temperature Rating	50 deg C								50							
	130 deg C								130							
Flange Class	UFM # 150									1						
	UFM PN 10/16									2						
	UFM PN 40									3						
	UFM Tril-clover End									6						
Output	Pulse											S				
	RS485											C				
	Modbus Wired											M				
	Modbus Wireless											W				
	GSM/GPRS											G				
	LoRaWan											L				
Power Supply	UFM Battery												B			
	24 VDC												V			
Matching Flange	MMF Skip														O	
	MMF Mating Flanges														P	
Data Logger	MM Data Logger															P

Contact Us
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